

Game 07 — Spirit Runner (Complete Plan)

Game 7 of 7 Complexity: ★★ ★ High Estimate: 5–7 days Status: Not started

Complete plan (supersedes the 2026-09-09 sample). Status: **built 2026-09-11** — Phases A–D complete in `apps/frontend/public/games/spirit-runner/` (`engine.js` forked from Game 05 with slide + swipe grammar, `gates.js` , `core.js` , `render.js` , `main.js`); §8 jest suite (`src/__tests__/games-spirit-runner.test.ts`) and the folder's `core.test.html` harness green. Outstanding: balance pass (§9 Phase D) and the 10-minute manual session (master README §5); P3 deferred below.

1. Overview

Mystical-forest endless runner with light puzzle choices: run, dodge, collect spirit orbs for powers, and at path splits pick the right **rune gate** — wrong pick banishes you to the riskier, richer **shadow realm**. Depth from choices, not controls (one thumb).

2. Complete game spec

Phase A — core run (fork of Game 05)

- Jump (as 05: coyote/buffer/cut) **plus slide**: swipe down / ↓ / S — 600 ms, hitbox drops to 40 % height; slide-jump cancels into a hop.
- Obstacles:

Type	Clearance	Look (procedural)	Introduced
Fallen log	jump	brown rounded rect	0 m
Low branch	slide	green bar at head height	300 m
Spirit guardian (tall)	jump only	glowing wisp column	800 m
Spirit guardian (low)	slide only	hovering wisp disc	800 m
Rune trap	time it	floor spikes pulsing: 1.2 s cycle — 0.7 s dark (safe), 0.5 s lit (lethal)	1200 m

- Speed ramp as 05 (320→900 px/s). **Hearts: 2**; hit = −1 heart + 1.2 s invulnerability (blink); 0 → game over (deliberately softer than 05 — longer-form run).
- Palette shifts by depth tier (dawn → dusk → night forest).

Phase B — orbs & powers

- Orbs spawn in arcs/lines riding the spawner (arc of 5–7 over a jump, line of 4 on flat). Meter: 10 orbs → full. Power = next in a **fixed cycle** (readable): Double jump → Dash → Slow time (durations in the table below).

Power	Duration	Effect
Double jump	8 s	second mid-air jump
Dash	1.5 s	invulnerable, destroys the next obstacle hit (exactly one)
Slow time	4 s	world ×0.6 (audio pitch drops too)

- Power button (bottom-right, thumb reach) lights when meter full; keyboard E; optional auto-trigger setting.

Phase C — rune gates & shadow realm (the twist)

- Every **600 m ± 100** the path splits into two gates; a hint banner shows 2 s before. `gates.js` (pure) defines **5 rules**:

Rule	Hint text	Correct gate
Parity	"Even spirits walk the left path."	side matching <code>orbsSinceLastGate % 2</code>
Shape echo	"Follow the doubled symbol."	gate whose rune == the previous gate's rune
Negation	"The moon door lies."	NOT the gate matching the banner's shape hint
Sequence	"▲▲▲..."	gate continuing the shown 3-symbol pattern
Color trail	"Trust the last light you caught."	side matching last orb's color

- API: `makeGate(ruleId, rng, runState) → { correctSide, hintText, runes }`; `resolveChoice(gate, side) → 'correct' | 'shadow'`. Rules cycle in order per run (learnable), order shuffles per run via seed.
- **Correct** → +100 points, continue forest. **Wrong** → **shadow realm**: 45 s, darker palette, obstacle density ×1.5, orbs worth ×2, guaranteed **+1 spirit shard** on survival; timer HUD countdown; auto-returns to forest (no soft-lock).

Phase D — characters & meta

- **Spirit shards**: `floor(meters/1000) + shadowSurvives + floor(correctGates/3)` per run.
- **Characters** (pick on menu; modifiers):

Character	Unlock	Modifier
Forest Spirit	start	power durations +50 %
Hunter	5 shards	every run starts with Dash charged
Monk	12 shards	Slow time also ×0.7 obstacle spawn rate

- Save persists shards + unlocks forever (best scores separate).

3. Screens & UI

State	Elements
<code>menu</code>	Title, character cards (locked show shard cost), best distance/shards,  Run, mute

State	Elements
playing	World + HUD: hearts, orbs → power meter + button, distance, depth tier
gate	In-world split + hint banner (run continues — no separate screen)
shadow	Palette swap + "SHADOW REALM" banner + shard countdown
paused	Overlay (auto on visibilitychange)
gameover	Distance, orbs, gates correct, shards earned, unlocks, [↺ Retry] [Character] [🔗 Share]

Virtual viewport 900×500 like 05; `?debug=1` draws hitboxes + gate rule id (dev aid).

4. Controls

- Jump: tap / Space / ↑ (double jump while empowered). Slide: swipe down / ↓ / S.
- Power: button (bottom-right) / E; auto-trigger accessibility setting.
- Swipe-up = jump, swipe-down = slide (pointer gesture thresholds: 30 px, 120 ms).
- **Gate choice (built 2026-09-11, plan left it open):** while a split is live the outer thirds of the screen are the doors — tap the left third for the left door, right third for the right (← / → on keyboard); the middle third keeps jumping and slides still work. While the doors are on the track the spawner holds and the spawn cursor jumps past them, so no obstacle ever overlaps the choice window; a split answered by nothing (the doors pass the runner) resolves as `shadow` — the realm takes you, never a soft-lock (\$7.7).

5. Data model (localStorage)

```
game:sprite-runner:save → {
  bestDistanceM: 2340, shards: 17, unlocked: ['spirit'],
  character: 'spirit', settings: { autoPower: false, muted: false }
}
```

6. File structure & function inventory

```
spirit-runner/
  index.html
  style.css
  engine.js  # forked from 05: loop, camera, parallax, input (jump+slide+swipes)
  core.js    # pure: obstacles, orbs, powers, gates resolution, shards – test surface
  gates.js   # pure: the 5 rune rules + hint text
  render.js  # palettes (forest/shadow), glow (lighter composite), particles, HUD
  main.js    # state machine + meta (characters, unlocks)
```

`gates.js` : `RULES` array, `makeGate(ruleId, rng, runState)`, `resolveChoice(gate, side)`,
`hintFor(ruleId, runState)`. `core.js` : `stepPlayer` (jump/slide states), spawner

extensions (spawnOrbArc , spawnTrap , spawnGuardian), applyPower(run, kind) ,
tickPowers(run, dt) (timers incl. character modifiers), shardsFor(run) .

7. Edge cases

1. Gate while a power is active → power timers keep running through the split.
2. Wrong-gate during invulnerability → shadow realm still triggers (it's not damage).
3. Dash inside shadow realm destroys one obstacle only, as in forest.
4. Slow time stacks with Monk modifier → world $\times 0.6$, spawn $\times 0.7$ (multiplicative, capped once each — no double application).
5. Shadow realm timer always expires exactly (fixed-step accumulation; pause-safe); pays shards exactly once.
6. Unlocks persist across character switching; switching mid-menu only (never mid-run).
7. Rule sequence: 5 rules cycle; run seed shuffles the order; no rule repeats within a cycle; gates ≥ 600 m apart never overlap the shadow realm window (spawner defers).
8. Storage disabled → unlocks lost per session, playable.

8. Testing plan (gates.js + core.js)

- All 5 rules: makeGate + resolveChoice → correct side deterministic given a fixed runState (table tests per rule, 20 rng seeds each).
- Hint text present and never empty for any rule/state.
- Shard math: boundaries at 999/1000 m, 1/2/3 correct gates, shadow survive.
- Power timers: durations incl. Forest Spirit $\times 1.5$; Dash single-destroy assertion.
- Shadow realm: density multiplier applied; timer expiry; single shard payout.
- Spawner: gate deferral rule (§7.7) over 1,000 simulated runs.

9. Task breakdown

Phase A — core run (≈ 2 days)

- ☒ Fork engine from 05; slide + low-branch/log/guardian/trap obstacles; trap pulse cycle
- ☒ 2 hearts + invulnerability; depth palettes; pause/blur; menu/gameover; best distance

Phase B — orbs & powers (≈ 1.5 days)

- ☒ Orb arcs in spawner; power meter + cycle; DJ/Dash/Slow implementations
- ☒ Power button + auto-trigger setting; WebAudio (collect/power/hit) + mute

Phase C — rune gates & shadow realm (≈ 2 days)

- ☒ gates.js 5 rules + hints (+ tests); split-path rendering; resolution flow
- ☒ Shadow realm: palette, density $\times 1.5$, $\times 2$ orbs, shard payout, timed return, banner

Phase D — characters & meta (≈ 1.5 days)

- ☒ Shard earning; unlock thresholds; 3 character modifiers; picker UI

- ☐ Balance pass: median first-run 30–45 s, each gate rule learnable ≤ 3 exposures (analytically tuned — obstacle intro at 0/300/800/1200 m, splits 600 ± 100 m, hearts 2 — but no human playtest yet)
- ☒ Share (`Ran {m} m as {character}... {pts}`); QA gate (master README §5) — automated parts done (§8 suites green, offline static, isolation greps clean); the 10-minute manual phone-width session remains for the owner

P3 — polish

- ☐ Sprite/hand-drawn art slot (post-MVP); music track; daily depth seed; extra rules
- ☐ Achievements/leaderboards: only if isolation decision is reversed (master §7)

10. Acceptance criteria

- Each phase ships playable alone (A = good runner; A+B = powered; +C = full twist; D = meta).
- §8 tests green — all gate rules proven deterministic per state.
- Shadow realm never soft-locks; pays exactly once; always exits.
- 60 fps mid-range phone with particles + glow on; pause never desyncs timers.
- One-thumb: jump, slide, power reachable at 360 px width.

11. Deferred coupling (intentionally NOT built)

Footer/nav links, analytics events, platform achievements integration, leaderboards — per master README §7 (revisit only if the owner reverses isolation).

12. Rev 2 architecture upgrade (2026-09-11) — reference: `03-sliding-puzzle.md`

Owner-approved architecture reference for all games (Sliding Puzzle Rev 2), applied per this game's nature: spirit runner already has seeded, choice-rich runs (the run seed shuffles gate-rule order), so async "same seed" challenges compare runs fairly; the upgrade is persistence hygiene + config extraction. Isolation contract unchanged.

Target additions

```
spirit-runner/
  index.html
  style.css
  engine.js      # unchanged
  core.js        # unchanged (obstacles, orbs, powers, shards – test surface)
  gates.js       # unchanged (5 rune rules)
  render.js      # unchanged
  main.js        # unchanged
  config.js      # flags + per-locale strings (share template; gate hints stay in
                  # gates.js – rule-coupled), host-overridable; §2 tuning constants stay
                  # in core.js (moving them is deferred until the balance settles)
  storage.js     # guarded facade: versioned `game:spirit-runner:save` (shards, unlocks,
                  # character, settings) + migrations + remote adapter slot (created
                  # 2026-09-12, extracted from main.js)
```

Why versioning matters here

This is the composite save most likely to need a migration later (shards, unlocks, character, settings in one document) — version it **before** players accumulate shards, and migrate on read.

config.js

The §2 tuning tables (power durations, densities, shard formula) remain the spec of record; `config.js` is where the shipped values live so balancing passes (§9 Phase D) are data edits. Per-locale strings for gate hints, share template, character copy. Host override `window.__SPIRIT_RUNNER_CONFIG__` → `?locale=` → defaults. `prefs` may override `config` but is a cache, not the source of truth.

Multiplayer (per this game's nature)

- **Async seeded run — the natural fit:** share the run seed → identical gate-rule order and spawner stream; compare distance/shards. Gate choices stay the human skill, so the comparison is fair without a server.
- **Hot-seat:** alternate runs on one device, compare shards/distance — trivial.
- **Live co-op/race (incl. shadow realm): gated** — backend + server timing + §7 reversal. Not planned.

Phases

- ☒ R2-1 Persistence hygiene (2026-09-11; completed 2026-09-12): `config.js` (locale + share template + the host-injectable `remoteAdapter` seam; host-overridable), the versioned save extracted to `storage.js` (writes carry `version: 1`; load normalizes per field), and the tall-guardian gap fixed per §2 (spawner emits `h = GUARDIAN_TALL_H`; `obstacleBox` full-height jump-only branch). **Deferred:** moving the §2 tuning constants into `config.js` — they live in `core.js` and are imported by every module; a dedicated pass when the balance settles. Gate-hint copy stays in `gates.js` (rule-coupled). 43/43 tests green.
- ☐ R2-2 Async seeded-run challenge (`?seed=` + share template carrying it).

- ☐ R2-3 Still owed from §9 Phase D: the balance pass and the owner's 10-minute manual session — unchanged by Rev 2.

Source: plan/games markdown · generated 2026-09-09 · part of the 2D Games sample plans (plan/games/pdf).